

TRAUMATIC HEAD INJURY

Introduction

- A condition in which there is an hit to the brain tissue by an external force that lead to impairment at different levels.
- Impairment can occur at the level of
 - Consciousness
 - Behavior
 - Emotion
 - Physical
 - Cognitive
- Main etiologies of the head injuries are
 - RTA
 - Gunshot Injury
 - Riots and assaults
 - Fall from height



Types of Traumatic Brain Injury

- **Primary Injury**

- Due to brain tissue coming into contact with any object directly or acceleration/deceleration of the brain
- Direct hit often result in Contusion, Laceration and Intracerebral hematoma
- Acceleration/Deceleration can cause compression force on the brain that can further lead to **Diffuse Axonal Injury**
- Can be seen in high speed RTA and some sport injury
- DAI can lead to Wallerian type of axonal degeneration

- **Secondary Injury**

- Death of brain cells due to a series of cellular events resulting from Hypoxia, Ischemia, Elevated ICP etc

Types of Traumatic Brain Injury

- Closed Head Injury
 - The cranium remains intact after the injury
 - The brain tissue gets damaged due to inertial forces on the brain when the skull got a powerful blow.
 - If the brain injury occurs on the same side of the blow over skull, then it is called Coup Injury
 - If the brain injury arises on the opposite side, then it is called Countercoup Injury
 - Closed Head Injury is further classified into different types like
 - Concussion
 - Cerebral Hematoma
 - Diffuse Axonal Injury
 - Intracranial Hemorrhage
 - Intra-axial hemorrhage
 - Extra-axial hemorrhage – Epidural & Subdural Hemorrhage
 - Subarachnoid Hemorrhage

Closed Head Injury

○ Concussion

- Mildest form of injury
- Due to any blow to the head it causes the head or the brain to move rapidly back and forth
- It can lead to physical, cognitive and emotional disturbances
- Severity is measured by concussion grading system
 - Grade 1 – Confusion only
 - Grade 2 – Confusion and Post-traumatic amnesia
 - Grade 3 – Confusion, Post-traumatic amnesia and physical; disturbance
 - Grade 4 – Loss of consciousness
- Patient can also lead to PTSD
- In some cases antegrade and retrograde amnesia can also be seen
- The duration of amnesia correlates with the severity of amnesia

Closed Head Injury

○ Cerebral contusion

- Bruising of the brain tissue
- There will be scattered areas of bleeding on the surface of the brain
- In majority of cases, contusion occurs in frontal and temporal lobes
- It can be extremely dangerous and life threatening
- Cerebral edema and trans tentorial herniation can also be seen in the severe condition
- Trans tentorial herniation is the movement of brain tissue from one compartment to other
- Immediate surgical intervention may be required

Closed Head Injury

○ Diffuse axonal injury

- Usually due to acceleration or deceleration motion
- Axons are stretched and damaged widely due to slide over another
- Prognosis varies depending on the extent of damage
- It results in instant loss of conscious

○ Intracranial hemorrhage

- The arteries in the cranium can be ruptured due to injury and can lead to hemorrhage
- It is considered as focal brain injury
- It occurs at a localized spot rather than causing the diffused spread
- This is of two types
 1. Intra-axial hemorrhage
 2. Extra-axial hemorrhage

Closed Head Injury

○ Intra axial hemorrhage

- Bleeding within the brain itself
- Also known as Cerebral Hemorrhage
- The bleeding can be in the brain tissues or the brain ventricles
- More dangerous and harder to treat

○ Extra axial hemorrhage

- Bleeding that occurs within the skull but outside the brain tissue
- It is classified into
 1. Epidural Hemorrhage
 2. Subdural Hemorrhage
 3. Subarachnoid Hemorrhage

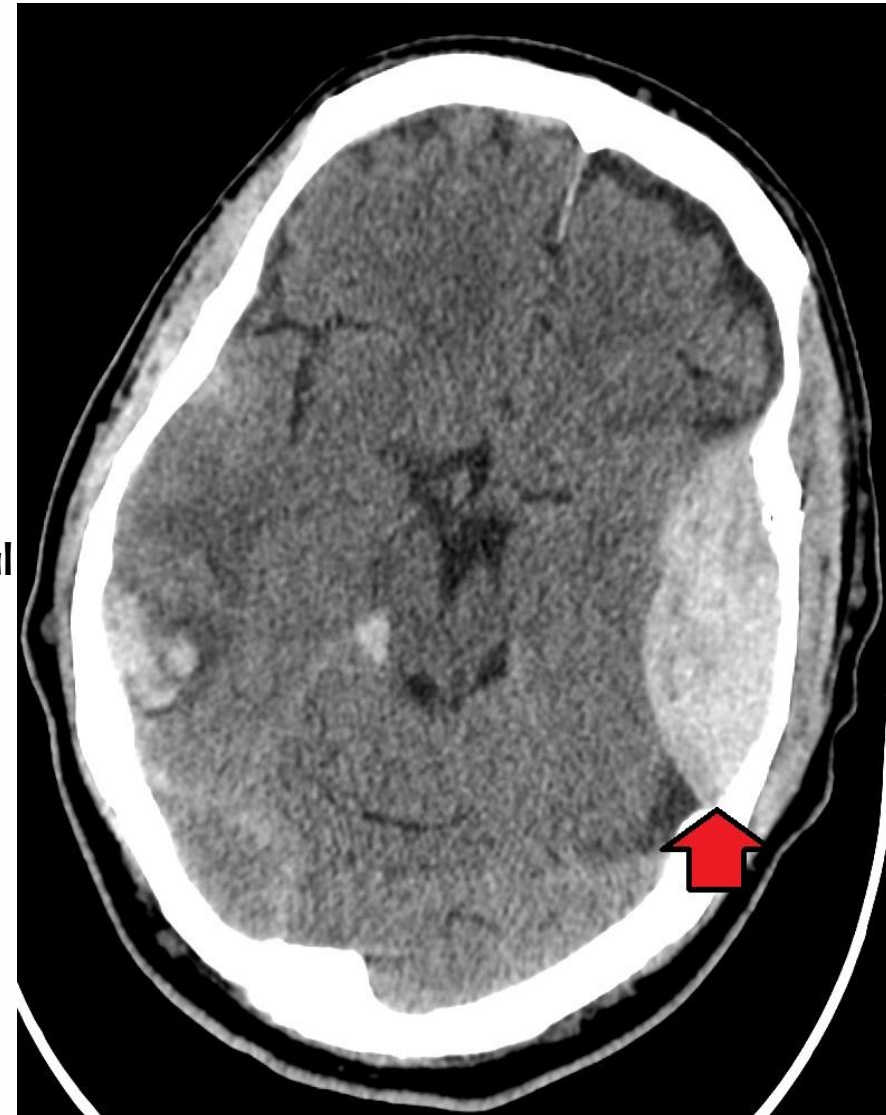
Closed Head Injury

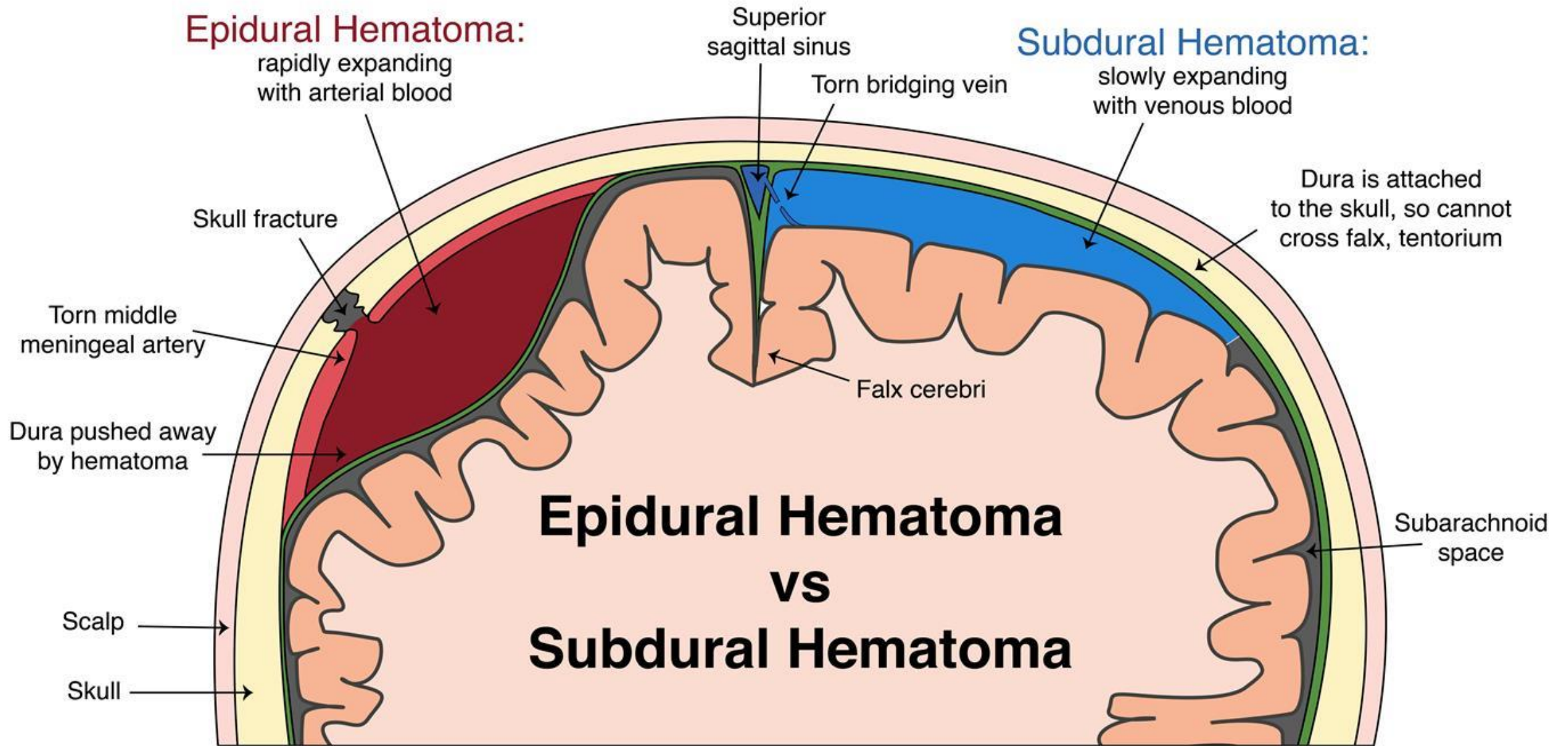
○ Epidural hemorrhage

- Also known as Extradural Hemorrhage
- Hemorrhage occurs between dura matter and the skull
- Mainly because of the laceration of an artery due to trauma
- Middle meningeal artery is commonly involved

○ Subdural hemorrhage

- Bleeding due to the rupture of the bridging veins in the subdural
- Between dura matter and arachnoid matter

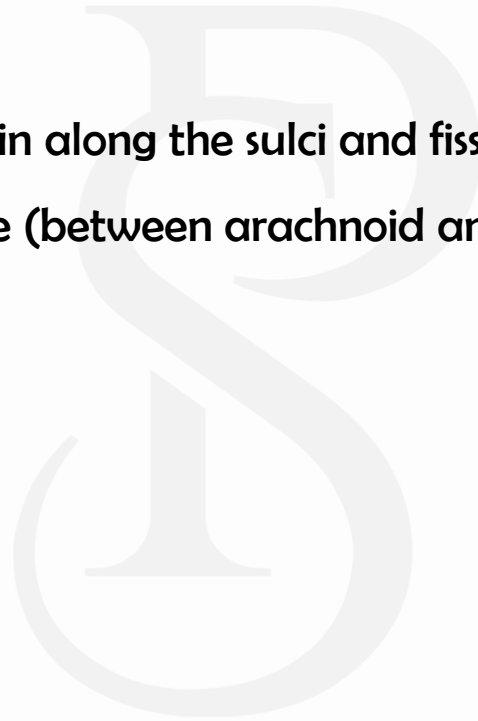




Closed Head Injury

○ Subarachnoid hemorrhage

- Due to trauma or aneurysm
- Blood starts to layer into the brain along the sulci and fissures
- Occurs in the subarachnoid space (between arachnoid and pia matter)



Clinical Manifestation of TBI

Clinical Manifestations of Head Injury

○ Altered state of consciousness

- There can be some dysfunction in the state of consciousness of the patient
- Mainly due to B/L cerebral hemisphere dysfunction or direct depression to the brainstem
- Various altered state of consciousness can be seen in the patient

1. Coma

- Complete arrest of all cerebral functions
- A period of prolonged unconsciousness to any type of stimuli

2. Stupor

- State of unresponsiveness but responds to painful stimulus through body movements

3. Delirium

- State of disorientation
- Stimulus is misinterpreted

Clinical Manifestations of Head Injury

○ Cognitive dysfunction

1. Attention deficits

- Patient will be unable to maintain attention

2. Memory loss

- Patient may have temporary or permanent memory loss
- 3 forms of memory loss can be seen generally
 - **Antegrade amnesia** – Inability to store new memory after the injury
 - **Retrograde amnesia** – Inability of the patient to recollect specific memory that took place in the past before the trauma
 - **Post-traumatic amnesia** - Loss of the memory between the time of injury and recovery

Clinical Manifestations of Head Injury

○ Cognitive dysfunction

3. Speed of processing

- Patient with frontal lobe trauma will have problem in the speed of processing and executing the functions
- Patient will be unable to
 - Choose a goal
 - Develop the plans to reach the goals

4. Emotional changes

- If orbitofrontal area is involved in trauma, patient will have
 - Euphoria
 - Intolerance
 - Irritability
 - Inappropriate sexual and social behaviour

Clinical Manifestations of Head Injury

○ Cognitive dysfunction

5. Behavioral changes

- Patient may show deviation from the normal behaviour
- Following symptoms can be seen
 - Depression
 - Impulsiveness
 - Hyperactivity
 - Involuntary expression of crying or laughing

Clinical Manifestations of Head Injury

○ Speech and language disorder

1. Broca's aphasia

- Patient will have difficulty in speaking
- Speech is broken, unclear or sometimes absent completely
- Also called “Motor Aphasia”

2. Wernicke's aphasia

- Patient will not understand the spoken words
- Patient will speak complete sentence fluently
- Also called fluent or sensory aphasia

3. Global aphasia

- Both the ability (speaking and understanding) will be lost
- Due to severe brain injury

Clinical Manifestations of Head Injury

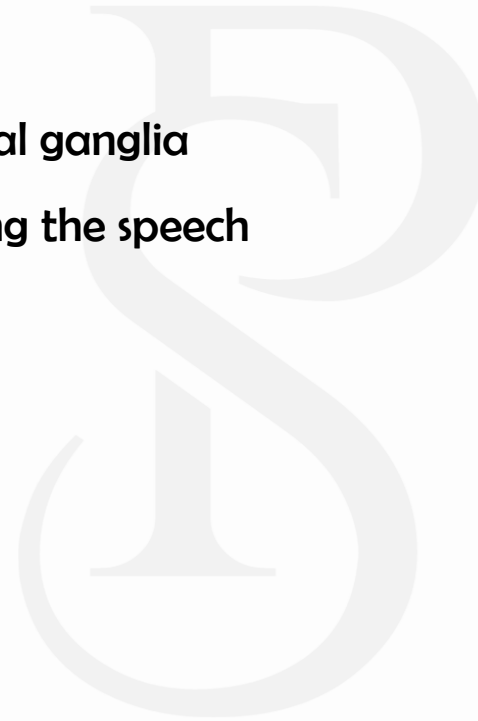
- Speech and language disorder

- 4. **Dysarthria**

- Seen in lesion with basal ganglia
 - Difficulty in articulating the speech

- Sensory dysfunction

- Motor dysfunction



Clinical Manifestations of Head Injury

- Sensory dysfunction
- Motor dysfunction



Clinical Manifestations of Head Injury

- Sensory dysfunction
- Motor dysfunction



PT Assessment for TBI

Assessment

○ History

1. Present History

- Time and Date of injury
- Location of injury
- Type or Mechanism of injury
- First aid
- Technique of carrying the patient
- Duration of unconsciousness
- External bleeding

2. Past History

- Any medical conditions
- Any trauma in past

Assessment

○ History

2. Past History

- Any current medications or any past medications
- Any pathological condition

3. Surgical History

- Any surgery in the past
- If yes, then which type of surgery
- Location of surgery
- Date of surgery

4. Family History or H/O of genetic disease

Assessment

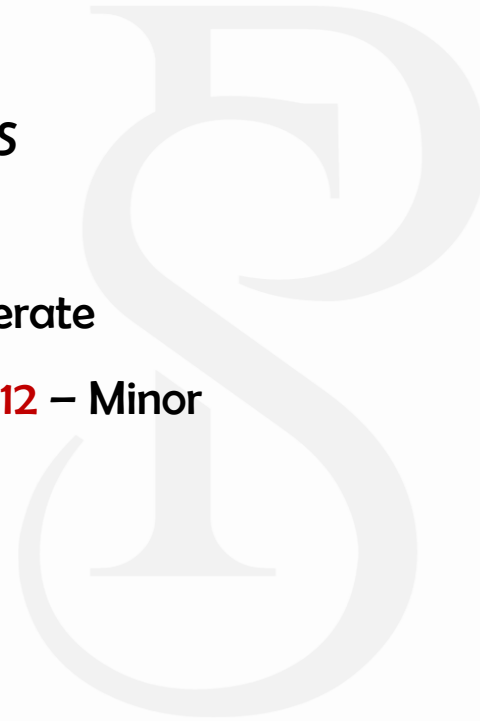
○ Observation

- General condition of patient
- Any abnormal posture of patient
- If the patient is on artificial ventilation, note the parameters
- Observe various tubing like catheter, Ryles tube etc
- Skin color
- Site and extent of injury
- Bruises in the body
- Observe the swelling
- Any scar

Assessment

○ Examination

- Level of consciousness
 - Can be assessed by GCS
 - **Below 8-9 – Severe**
 - **Between 9 – 12 – Moderate**
 - **More than or equal to 12 – Minor**



Glasgow Coma Scale

○ Eye Opening Response

- Spontaneous – 4
- To loud voice – 3
- To pain – 2
- None – 1

○ Verbal Response

- Oriented – 5
- Confused/disoriented – 4
- Inappropriate words – 3
- Incomprehensive sound – 2
- None – 1



Glasgow Coma Scale

○ Motor response

- Obeyes – 6
- Localizes – 5
- Flex to withdraw from pain – 4
- Abnormal flexion posture – 3
- Abnormal extension – 2
- None – 1



Assessment

○ Examination

- Cranial nerve examination
- Sensory examination
- Reflex examination
- Muscle tone
- MMT
- ROM
- Coordination
- Balance
- Synergy patterns
- Gait assessment



PT Management for TBI

(Unconscious Patient)

Unconscious Patient

○ Aim of management

- To stimulate arousal response in the patient
- To maintain good bronchial hygiene
- To prevent secondary complications
- To maintain joint ROM

○ Precautions

- Monitor vital signs
- Physio should be aware of the effects of the positioning and mobility
- Frequency and intensity of the treatment should be determined as per the patient
- Physiotherapist must be aware about the patient's medications, its adverse effects, lab investigations, radiological evaluation, physical treatment by other team.

Unconscious Patient

○ Management

- Postural drainage
- Chest secretions may be mobilized by manual techniques like percussions and vibration.
- Head low position must be avoided in patients with head injuries due to risk of increasing ICP
- Stretching of pectorals and sternocleidomastoid can help to maintain chest mobility
- Neurophysiological Stimulation of respiration
- **Perioral technique** – Moderate finger pressure is maintained inwards and downwards just above the lip.
- **Intercostal stretch** – Apply pressure downwards towards the toes on the upper border of the rib at the end expiration (not on floating ribs)
- **Co-contraction of abdominal muscles** – Apply pressure laterally over the lower ribs and pelvis at right angles to patient and maintain the pressure for 2 minutes

Unconscious Patient

○ Management

- **Vertebral pressure** – Finger pressure against thoracic vertebrae between T2 and T10
- **To avoid contracture and deformities**
 - Passive ROM of all joints
 - Prolonged stretching
 - Splints should be used
 - Proper positioning
 - Change of positioning
- **To avoid cardiorespiratory complication**
 - Passive movements
 - Elastic crepe bandage
 - Limb elevation
 - Gentle massage techniques

Unconscious Patient

○ Management

- To avoid cardiorespiratory complication
 - Proper positioning on bed
 - Postural drainage
 - Use manual techniques like percussions and vibration.
 - Best position for relaxation and improved ventilation & oxygenation are with head to the bed elevated to 30 degrees
 - Leg lifts
 - Stretching
- To maintain good skin integrity and prevent pressure sores
 - Closely monitor all the \ areas of high risk
 - Frequent change in position
 - Rubbing massage with moisturizer for a small duration over the skin

Unconscious Patient

○ Management

- To maintain good skin integrity and prevent pressure sores
 - Use of air bed
 - Avoid creases on bed
 - At redness stage, stop the massage and start changing positions more frequently
 - At blister stage, dressing should be done
- To maintain normal joint ROM
 - Stretching exercises
 - Orthotic support can be beneficial
 - Serial casting can be used (often used for plantar flexor or bicep contracture)
 - Strengthening exercises
 - Passive movement

PT Management for TBI

(Conscious Patient)

Conscious Psatient

○ Aim of management

- To provide psychological support
- To normalize the tone and improve muscle strength
- To improve voluntary control
- To train the transfers and early ambulation
- To enhance motor relearning and training
- To improve overall functional capabilities
- To prevent all possible secondary complications

Conscious Patient

○ Management

- To normalize muscle tone and muscle properties
 - Hypotonicity
 - Ice brisk stroking over the muscle
 - Hacking over the muscle
 - Weight bearing exercises
 - Aquatic exercise
 - Functional re-education exercises
 - MET
 - Hypertonicity
 - PNF Techniques
 - Stretching exercises
 - Orthotic support to maintain muscle strength

Conscious Patient

○ Management

- To improve voluntary control
 - Movement training should be goal oriented and improved through practice
 - PNF Techniques
 - Strengthening exercises
 - Biofeedback
 - Functional re-education exercises
 - Vestibular ball exercise will also promote the control over the voluntary movements
 - Repetition of the activities
 - While performing any movement, we have to focus on:
 - Purpose of the movement
 - Order of the movement
 - Direction

Conscious Patient

○ Management

- To improve voluntary control
 - While performing any movement, we have to focus on:
 - Rate of movement
 - Timing
 - Postural adjustments
 - Limbs movement
 - Coordination
 - To develop control over voluntary movement, normalizing the tone, improving strength, correct pattern of movement, coordination and normal timing should be taught.

Conscious Patient

○ Management

- To improve early ambulation and transfers
 - Assess the patient's ability for transfer activities.
 - Initially these are taught to and from the bed or the wheelchair
 - Weight transfers should be taught
 - Lying to sitting
 - Sitting to standing
 - Standing to sitting
 - Before starting the gait training, identify the impairments that primarily determine walking ability of the patient
 - Gait training protocol can be framed on the functional independence level of the patient

Conscious Patient

○ Management

- To improve early ambulation and transfers
 - Train for forward walking in parallel bars
 - Then progress to walking aids
 - Lastly progress it to independent walking
 - Side stepping
 - Reverse walking
 - Pivot turning
 - Stair climbing

Conscious Psatient

- Management

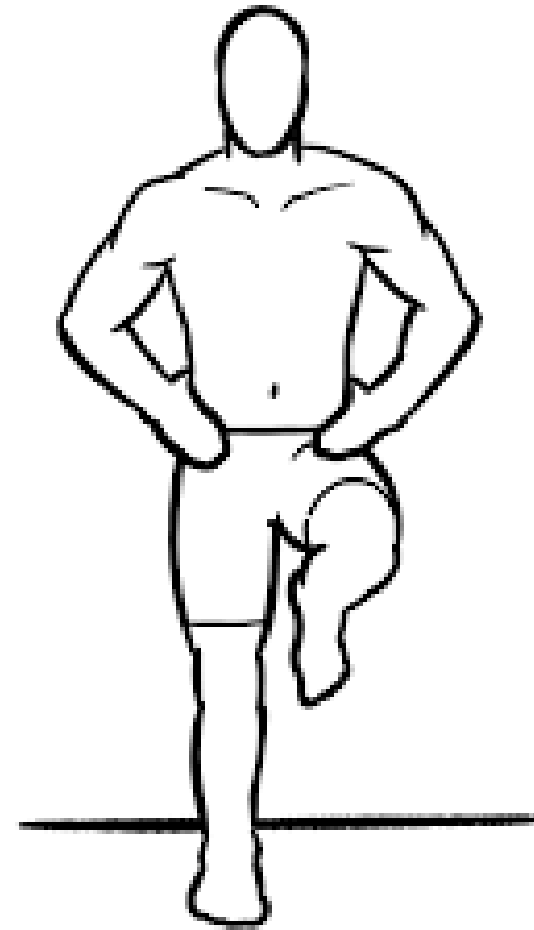
- To train for balance and equilibrium
 - Single limb standing



Conscious Psatient

○ Management

- To train for balance and equilibrium
 - Single limb standing with hip 90 degree flexion



Conscious Psatient

○ Management

- To train for balance and equilibrium
 - Balance board activities
 - Swiss ball activities
 - Toe standing
 - Toe walking



Conscious Patient

○ Management

- To train for balance and equilibrium

